

Project Initialization and Planning Phase

Date	26 June 2024
Team ID	-
Project Title	EduGenie – AI-Powered Educational Assistant
Maximum Marks	5 Marks

Project Proposal (Proposed Solution) report

The proposal aims to give students and self-learners a single AI-powered educational assistant that answers questions, explains concepts, generates quizzes, summarizes documents, builds learning roadmaps, and tracks progress — replacing scattered tools with one authenticated workspace.

Project Overview	
Objective	To help students learn more effectively by combining Google Gemini-powered AI tools with note-taking, bookmarking, and learning-analytics features in a single secure web platform.
Scope	The project covers a Fast API backend with JWT authentication, MongoDB Atlas storage, Gemini-based AI endpoints (Q&A, Explain, Quiz, Summarize, Roadmap), and a responsive HTML/CSS/JS frontend with dashboard, notes, bookmarks, history, progress, and settings pages.
Problem Statement	
Description	Students rely on fragmented tools — search engines for answers, separate apps for notes, and no automated way to summarize long material or track study progress — which slows down learning and revision.

Impact	Solving these issues will result in improved operational efficiency, reduced risks, and an overall enhancement in the lending process, contributing to customer satisfaction and organizational success.
Proposed Solution	
Intelligent Q&A;	Employing machine learning techniques to analyse and predict creditworthiness, creating a dynamic and adaptable loan approval system.
Key Features	Solving this reduces time spent switching between tools, improves concept retention through structured explanations and quizzes, and gives learners visibility into their own progress and consistency.
Quiz Generator	MCQ, True/False, Fill-in-blank, and Short-answer quizzes with adjustable difficulty
AI Summarization	Text Summarization
Learning Roadmap	Phased, milestone-based learning plans for any topic/level/duration
Smart Notes & Bookmarks	Folder-organized, autosaving notes and saved AI responses
Progress Dashboard	Streaks, focus score, sessions completed, quiz performance and weekly goals
Secure Auth	JWT-based registration/login with bcrypt password hashing

Resource Requirements

Resource Type	Description	Specification/Allocation
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Hardware		
Computing Resources	CPU/GPU specifications, number of cores	T4 GPU
Memory	RAM specifications	8 GB
Storage	Disk space for data, models, and logs	1 TB SSD
Software		
Frameworks	Backend Framework	Fast API
Libraries	Additional libraries	google-generative Ai, motor, pymongo, python-jose, passlib, bcrypt, PyPDF2, python-docx, python-multipart, aiofiles, HTTPs, slow Api, pydantic, pydantic-settings
Development Environment	IDE	Visual Studio Code, PyCharm
Database	Database Management System	MongoDB Atlas
Data		
Data	Source, Format	User-generated data, AI-generated responses, PDF/DOCX/TXT uploads, MongoDB collections (JSON/BSON)